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Student Number

2022

Preliminary Higher School Certificate Examination

Mathematics Standard

General Instructions

- Reading time — 10 minutes
- Working time — 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
70

Section I — 10 marks (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II — 60 marks (pages 9–23)

- Attempt Questions 11–25
- Allow about 1 hours and 45 minutes for this section

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Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 It is given that $s = ut + \frac{1}{2}at^2$.

What is the value of s when $u = 3.11$, $t = 4$ and $a = 4.52$, correct to two decimal places?

- A. 13.60
- B. 21.48
- C. 48.60
- D. 85.92

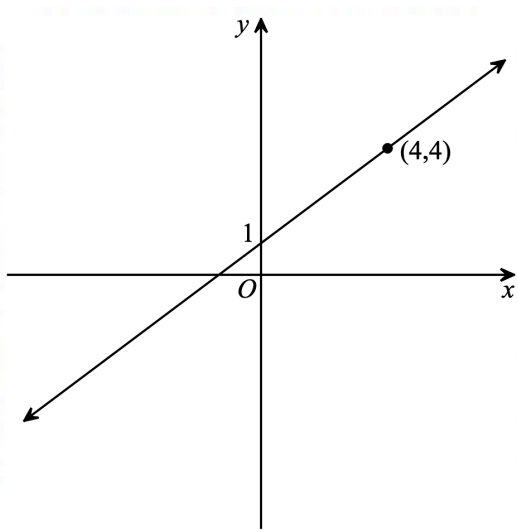
- 2 A pencil is measured to be 12.4 cm.

What is the percentage error in this measurement, correct to three decimal places?

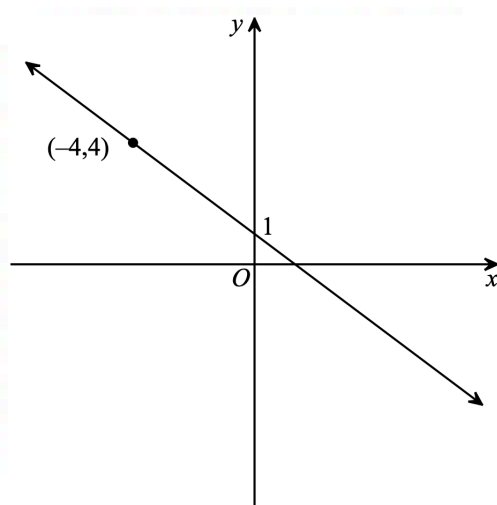
- A. 0.081%
- B. 0.403%
- C. 0.806%
- D. 4.031%

3 Which of the straight lines shown below represents the equation $y = \frac{3}{4}x + 1$?

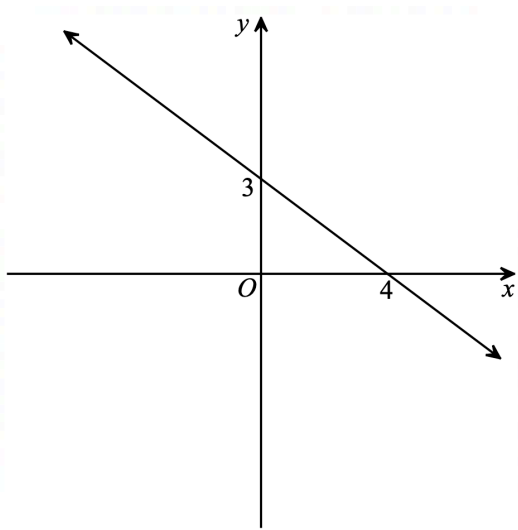
A.



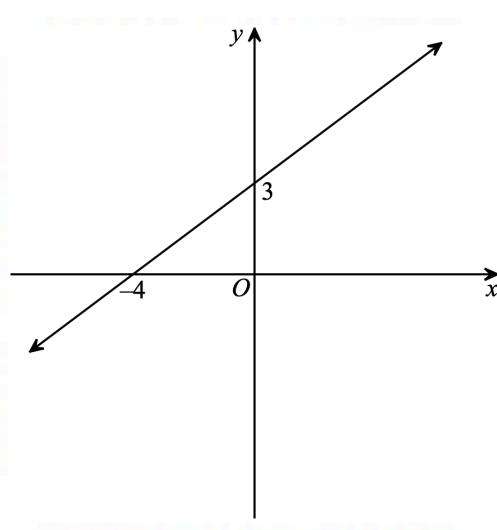
B.



C.

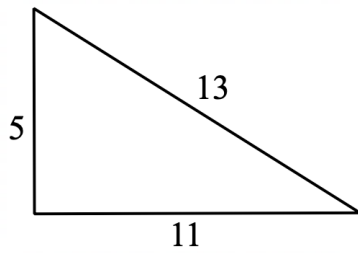


D.

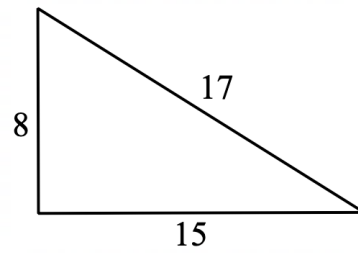


4 Which of the following triangles is right-angled?

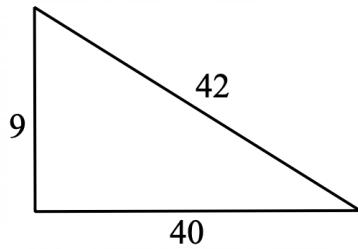
A.



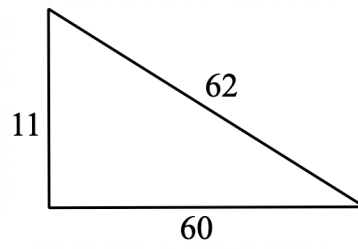
B.



C.



D.



5 A vase is sold at a store for \$16.50, including GST.

What is the cost price of the vase, before GST is added?

A. \$14.85

B. \$15.00

C. \$18.15

D. \$20.00

6 A concierge asks every 5th patron at a hotel to complete a feedback form.

What type of sampling has been used?

A. Random

B. Stratified

C. Systematic

D. Quantitative

- 7 Robbie has scored the least number of runs in his cricket team this season.

What will happen to the mean and the range of the team's scores if Robbie's score is removed from the data?

- A. The mean will decrease and the range will decrease
- B. The mean will decrease and the range will increase
- C. The mean will increase and the range will decrease
- D. The mean will increase and the range will increase

- 8 Last week, Claudia bought 18 eggs and found that 6 were broken when she got home. This week, she counts that there are 3000 eggs on the shelf at the same grocery store.

How many eggs on the shelf can Claudia expect to be broken?

- A. 1000
- B. 1200
- C. 1500
- D. 1800

- 9 Which of the following is equivalent to 100 m^2 ?

- A. 0.0001 km^2
- B. 0.001 km^2
- C. 0.01 km^2
- D. 0.1 km^2

- 10** How long does \$500 need to be invested at 7% per annum simple interest in order to double its value?
- A. 11 years
 - B. 13 years
 - C. 15 years
 - D. 17 years

End of Section I

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Student Number

Section II

60 marks

Attempt Questions 11 – 25

Allow about 1 hour and 45 minutes for this section

Instructions

- Write your Student Number at the top of this page.
 - Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided.
If you use this space, clearly indicate which question you are answering.
-

Please turn over

Question 11 (2 marks)

Clark’s formula, given below, is used to determine the dosage of medicine for children. **2**

$$\text{Dosage} = \frac{\text{weight in kg} \times \text{adult dosage}}{70}$$

For a particular medicine, the adult dosage is 500 mg and the correct dosage for a specific child is 110 mg.

How much does the child weigh, to the nearest kg?

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Question 12 (2 marks)

Solve $x + 4 = \frac{x - 2}{5}$. **2**

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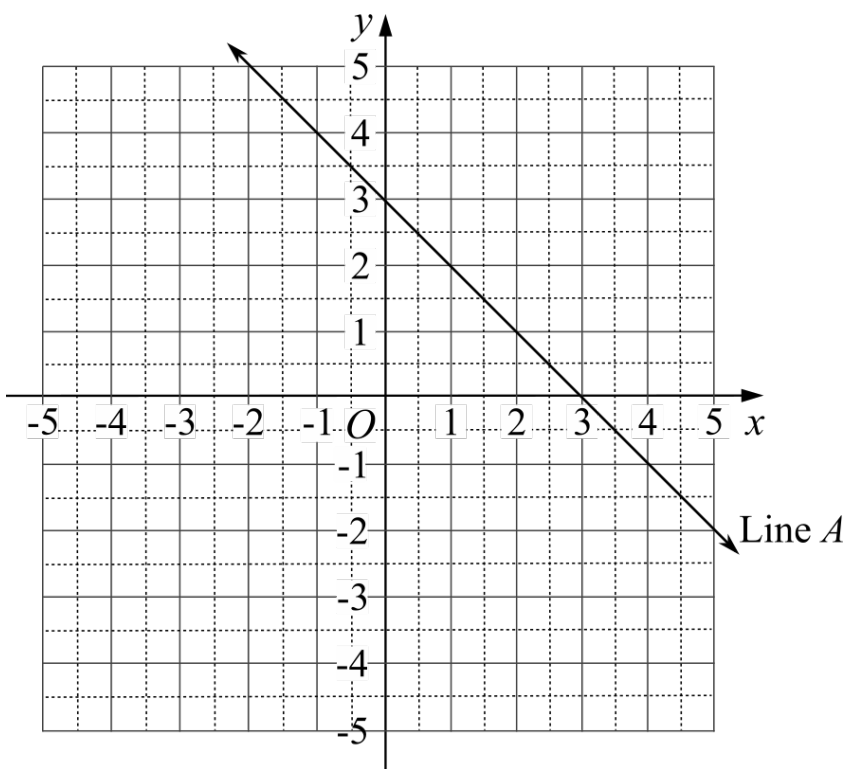
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Question 13 (4 marks)

The graph of Line A has been drawn on the number plane below.



(a) What is the gradient of Line A? 1

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(b) Draw a second line, Line B, that has equation $y = 3x - 3$. 2

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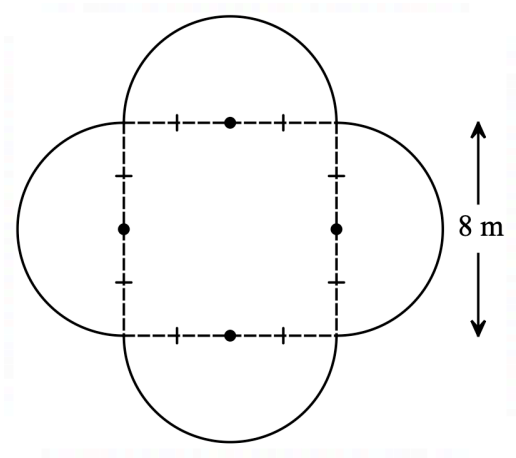
(c) Find the point of intersection between Line A and Line B. 1

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Question 14 (4 marks)

The diagram shows a garden enclosed by semi-circular arcs with diameter 8 metres.



- (a) Find the perimeter of the garden, correct to one decimal place. 2

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- (b) Find the area of the garden, correct to one decimal place. 2

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Question 15 (3 marks)

Jason is planning to build a new house and considers two different lighting options:

- Halogen light bulbs which consume electricity at a rate of 18 watts per hour and cost \$3.99 each.
- LED light bulbs which consume electricity at a rate of 10 watts per hour and cost \$14.95 each.

The average cost of electricity is 27c/kWh. Jason calculates that his house will need 35 light fittings which will be turned on for an average of 4 hours per day.

- (a) Find the difference between cost of purchasing LED light bulbs and Halogen light bulbs for all fittings. 1

- (b) How many days will it take for the total cost of purchasing and using LED light bulbs to equal the total cost of purchasing and using Halogen bulbs? 2

Question 16 (4 marks)

Liam lives in Berlin, Germany, located at $(52^{\circ}\text{N}, 13^{\circ}\text{E})$ and Kate lives in Santiago, Chile, located at $(36^{\circ}\text{S}, 72^{\circ}\text{W})$.

Every Sunday at 8pm Berlin time, Liam sends Kate an email with a photo of the moon.

- (a) Find the difference in the longitudinal coordinates of Liam and Kate. 1

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- (b) What is the day and time in Santiago when Kate receives the email? 3

It is given that $15^{\circ} = 1$ hour time difference. Ignore daylight saving.

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Question 17 (3 marks)

A brand new jet ski worth \$25 870 depreciates by \$700 every year after it is purchased. 3

In what year will the jet ski have a salvage value of \$20 000?

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Question 18 (5 marks)

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Jessica works as a traffic controller earning \$31.00 per hour for the first 35 hours worked each week, then double time for every hour thereafter. Her weekly budget and expenses are shown below.

	A	B	C	D
1	Jessica's weekly budget			
2				
3	Rent	\$350		
4	Utilities	\$60		
5	Travel expenses	\$95		
6	Personal expenses	\$150		
7	Food & Groceries	\$220		
8				
9	Total	\$875		
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Jessica is trying to save \$18 200 for a motorbike.

How many hours and minutes must she work on average each week to reach her savings goal in exactly one year's time?

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Question 19 (5 marks)

The table below shows the income tax payable for the current financial year.

Taxable income	Tax on this income
0 - \$18,200	Nil
\$18,201 - \$45,000	19 cents for each \$1 over \$18,200
\$45,001 - \$120,000	\$5,092 plus 32.5 cents for each \$1 over \$45,000
\$120,001 - \$180,000	\$29,467 plus 37 cents for each \$1 over \$120,000
\$180,000 and over	\$51,667 plus 45 cents for each \$1 over \$180,000

Stephen has a gross annual salary of \$113 000. His allowable tax deductions include \$3000 in business travel and \$6700 in self-education expenses. Stephen must also pay a Medicare Levy of 2% of his taxable income.

- (a) Calculate his taxable income.1

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- (b) Calculate the total tax payable by Stephen, including the Medicare levy.2

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- (c) If his employer has made fortnightly PAYG contributions of \$1216, find the tax refund Stephen will receive.2

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Question 20 (2 marks)

The table below shows the number of students in each grade at a local high school.

2

Year 7	Year 8	Year 9	Year 10	Year 11	Year 12
120	140	145	150	128	117

Sienna is reviewing the school’s uniform policy and intends to collect 200 responses using a stratified sampling technique.

How many surveys should be allocated to Year 10 students?

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Question 21 (5 marks)

The stem and leaf plot below shows the number of books read by Year 8 students during the most recent school holidays.

Stem	Leaf
0	2 3 4 6
1	0 1 2 2 5 9
2	2 3 7
3	1 1 2 6
4	7

- (a) Calculate the median number of books read.1

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- (b) Find the interquartile range for the data.2

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- (c) Determine whether there are any outliers in the data.2

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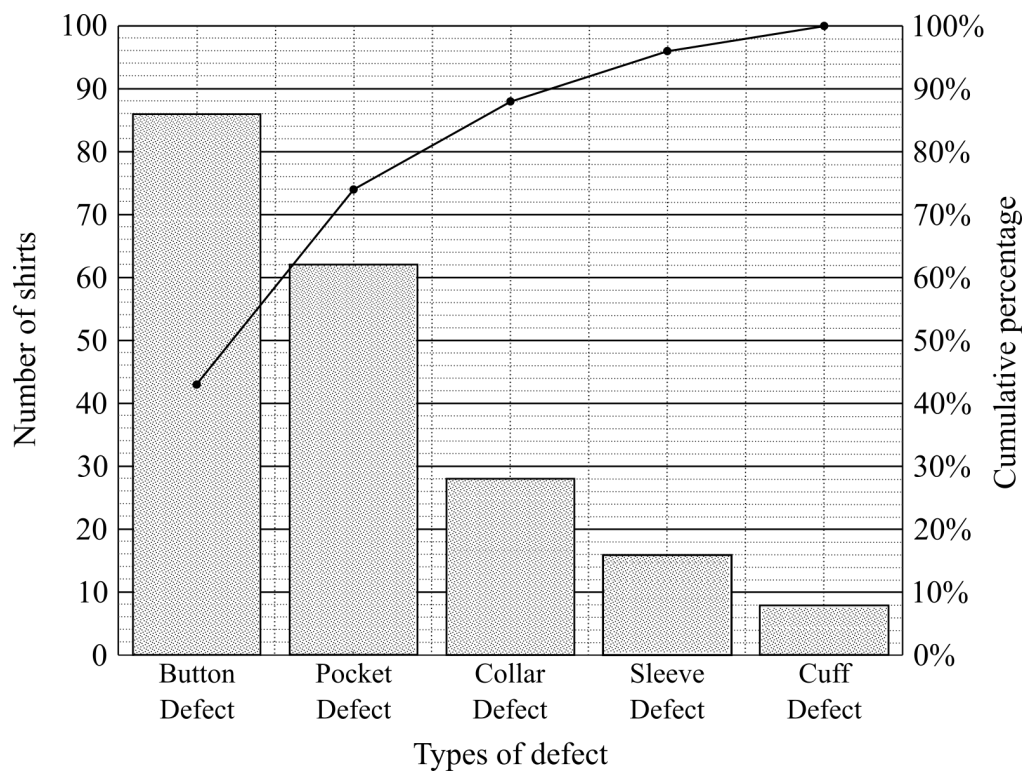
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Question 22 (3 marks)

The quality control team at the Shirts-R-Us factory collect shirts that are defective over a period of one week. They then sort them into the types of defect that each shirt has. The Pareto chart shows the results of the quality control team's findings.



(a) How many shirts were defective in total? 1

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(b) What percentage of shirts had a pocket or collar defect? 2

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Question 23 (4 marks)

An experiment is conducted where two dice are rolled and the sum of their uppermost faces are added.

- (a)

In the space provided below, use an appropriate diagram to show the possible outcomes of the experiment.

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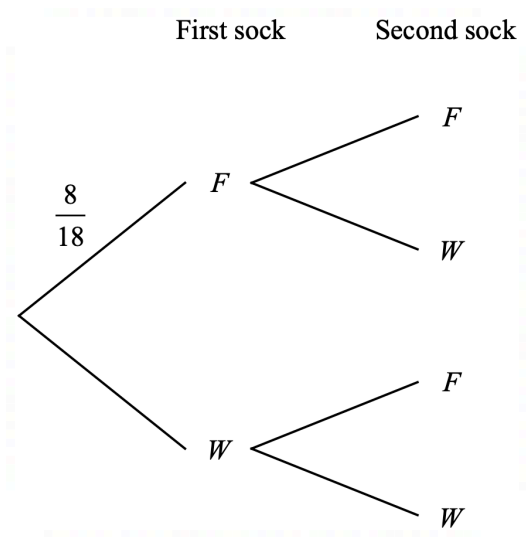
- (b)

Find the probability that the sum of the uppermost faces is greater than 6.

2

Question 24 (4 marks)

Brandon’s sock draw contains four pairs of football socks (F) and five pairs of workwear socks (W). Every morning he randomly selects two socks from the draw, without replacement. The partial tree diagram below shows the outcomes and probabilities.



- (a) Complete the tree diagram above

2
- (b) Find the probability that Brandon selects a matching pair of socks.

2

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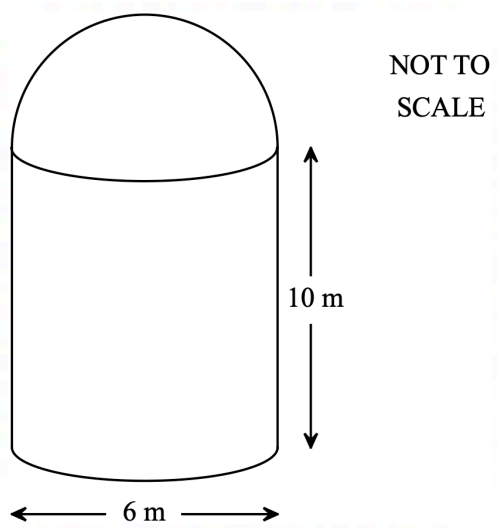
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Question 25 (10 marks)

A water tank is to be constructed using a cylindrical body with a hemispherical roof. The diameter of the tank measures 6 m and the height of the cylindrical body is 10 m, as shown on the diagram.



- (a) Find the surface area of the water tank, including the base, correct to two decimal places. 3

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- (b) The entire tank can be used to store water for a farm. 3

Find the capacity of the tank, correct to the nearest kL.

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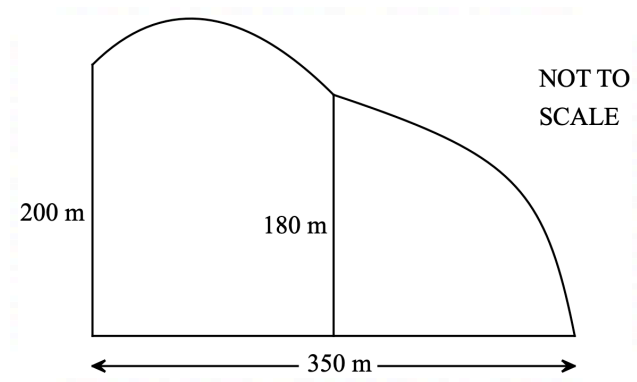
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Question 25 continues on page 23

Question 25 (continued)

(c) A map of the farm is shown below.



Using two applications of the trapezoidal rule, approximate the amount of water, in litres, that should be distributed evenly to each square metre of the farm. Give your answer correct to two decimal places.

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(d) Does the trapezoidal rule overestimate or underestimate the actual area of the farm? Justify your answer with reference to the diagram.

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End of Paper

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